

Estimation and Properties of Estimators

Maximum Likelihood Estimation (MLE)

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MAI 600: Mathematics and Statistics for AI

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Lecture Outline

The Core Problem: Estimation

A Motivating Example: Coin Flips

The Method of Maximum Likelihood

Worked Examples

Practical Considerations & Summary

The Estimation Problem: A Concrete Example

Scenario

Imagine we want to know the true average age of all students at a large university. Let's call this unknown value μ .

Surveying every student is impossible. Instead, we take a small random sample of 9 students and record their ages:

$$\{21, 24, 22, 47, 20, 18, 26, 19, 25\}$$

The Estimation Problem: A Concrete Example

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The Core Question

How can we use this sample to produce a single best guess for the true population mean, μ ?

Comparing Different Estimators

There's no single, obvious way to create an estimate. Here are three possible functions, or **estimators**, we could use.

Sorted data: {18, 19, 20, 21, **22**, 24, 25, 26, 47}

1. **Sample Mean:** $\hat{\theta}_1 = \frac{1}{9} \sum X_i = \frac{242}{9} \approx 26.89$
2. **Sample Median:** The middle value of the sorted data.
 $\hat{\theta}_2 = 22$
3. **Mid-Mean:** Average the three middle values.
 $\hat{\theta}_3 = \frac{21+22+24}{3} \approx 22.33$

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Which is Better?

- **Expected Value:** What is the average value of our estimator if we could repeat the sampling process infinitely?

$$\mathbb{E}[\hat{\theta}_1] = \mu \quad \mathbb{E}[\hat{\theta}_2] = \mu \quad \mathbb{E}[\hat{\theta}_3] = \mu$$

Comparing Different Estimators

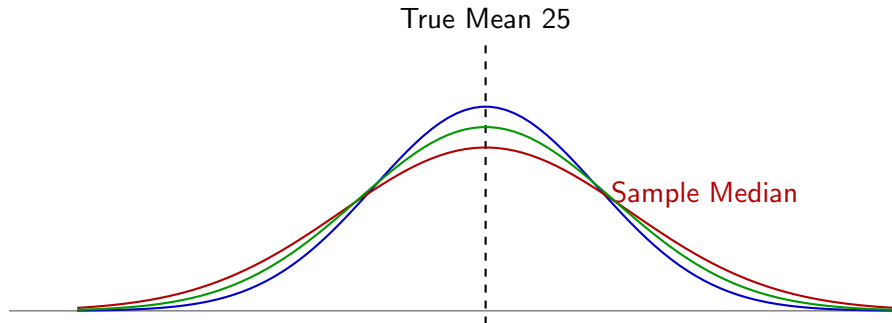


Figure: The best estimator is the one that is correct on average and has the smallest variance.

Evaluating Estimators

An **estimator** is a function of data used to estimate a parameter.
How do we measure its quality?

- **Bias:** Is the estimator correct on average?

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- ▶ **Mean Squared Error (MSE):** A single metric combining bias and variance.

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

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Suppose we flip a coin 10 times and observe 7 heads and 3 tails. We don't know if the coin is fair. What is our best estimate for the probability of heads, p ?

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The **Maximum Likelihood Estimation (MLE)** principle suggests we should choose the parameter p that makes our observed data *most likely*. Let $X_i = 1$ for a head, $X_i = 0$ for a tail. Our data is

$\{1, 1, 1, 1, 1, 1, 1, 0, 0, 0\}$.

- The probability of this specific sequence is:

$$L(p) = p \cdot p \cdot p \cdot p \cdot p \cdot p \cdot p \cdot (1-p) \cdot (1-p) \cdot (1-p) = p^7(1-p)^3$$

This function $L(p)$ is the **likelihood** of the parameter p .

Solving for the MLE

Our goal is to maximize $L(p) = p^7(1 - p)^3$. It's easier to maximize the **log-likelihood**, $\ell(p) = \log L(p)$, since the maximum will occur at the same point.

$$\ell(p) = \log(p^7(1 - p)^3) = 7 \log(p) + 3 \log(1 - p)$$

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Solving for p :

$$\frac{7}{p} = \frac{3}{1-p} \Rightarrow 7(1-p) = 3p \Rightarrow 7-7p = 3p \Rightarrow 7 = 10p$$

This gives the maximum likelihood estimate:

$$\hat{p}_{\text{MLE}} = \frac{7}{10} = 0.7$$

The sample proportion is the best estimate (intuitive).

How Good is Our Estimate?

Our coin flip estimator is $\hat{p} = \frac{\sum X_i}{n}$. Let us measure its quality.

► **Bias:** Our estimator is **unbiased** because

$$\mathbb{E}[\hat{p}] = \mathbb{E}\left[\frac{\sum X_i}{n}\right] = \frac{\sum \mathbb{E}[X_i]}{n} = \frac{np}{n} = p.$$

So, $\text{Bias}(\hat{p}) = 0$.

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- **Variance:**

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{\sum X_i}{n}\right) = \frac{1}{n^2} \sum \text{Var}(X_i) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}.$$

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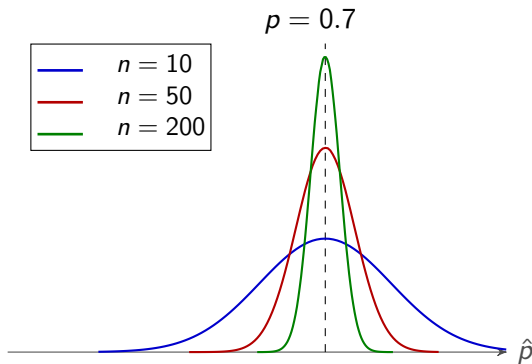
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- **Mean Squared Error (MSE):**

$$\text{MSE}(\hat{p}) = \text{Var}(\hat{p}) + \text{Bias}(\hat{p})^2 = \frac{p(1-p)}{n} + 0^2.$$

Illustration: The Estimator's Distribution

The variance $\frac{p(1-p)}{n}$ tells us that as the sample size n increases, our estimator $\hat{p}$ becomes more concentrated around the true value of p . This property is called **consistency**.



Fisher Information and the CRLB

Is there a theoretical limit to how good an unbiased estimator can be? Yes! The **Fisher Information**, $I_n(\theta)$, measures how much information the data $X_1, \dots, X_n$ provides about the parameter θ . It is defined as:

$$I_n(\theta) = \mathbb{E} \left[-\frac{\partial^2}{\partial \theta^2} \log L(\theta) \right] = n \cdot I_1(\theta)$$

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Cramér-Rao Lower Bound (CRLB)

For any unbiased estimator $\hat{\theta}$ of θ , its variance is bounded below:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)}$$

An estimator that achieves this bound is called **efficient**.

Back to the Coin Flip

Let's check if our estimator $\hat{p} = \bar{X}$ is efficient. The log-likelihood for one observation is $\ell_1(p) = x \log(p) + (1 - x) \log(1 - p)$.

- ▶ First derivative: $\frac{\partial \ell_1}{\partial p} = \frac{x}{p} - \frac{1-x}{1-p}$.
- ▶ Second derivative: $\frac{\partial^2 \ell_1}{\partial p^2} = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}$.

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Now, we find the Fisher Information for a single observation, $I_1(p)$:

$$\begin{aligned} I_1(p) &= \mathbb{E} \left[-\frac{\partial^2 \ell_1}{\partial p^2} \right] = \mathbb{E} \left[\frac{X}{p^2} + \frac{1-X}{(1-p)^2} \right] \\ &= \frac{\mathbb{E}[X]}{p^2} + \frac{1 - \mathbb{E}[X]}{(1-p)^2} = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} \\ &= \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)} \end{aligned}$$

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The CRLB for n samples is $\frac{1}{I_n(p)} = \frac{1}{n \cdot I_1(p)} = \frac{p(1-p)}{n}$.

Conclusion: $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$, which exactly matches the CRLB.

Our estimator is efficient!

Choosing Between Estimators

Scenario

We want to estimate the mean age of a population. Let's assume the true mean is $\mu = 40$ and the variance is $\sigma^2 = 150$. Which estimator is better?

- ▶ **Estimator 1:** Unbiased sample mean from $n = 10$ people.

$$\hat{\theta}_1 = \bar{X}_{10}$$

- ▶ **Estimator 2:** A slightly biased estimator from a huge sample of $n = 1000$ people.

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Let's compare them using MSE:

- $\text{MSE}(\hat{\theta}_1) = \text{Var}(\hat{\theta}_1) + \text{Bias}(\hat{\theta}_1)^2 = \frac{\sigma^2}{10} + 0^2 = \frac{150}{10} = 15.$

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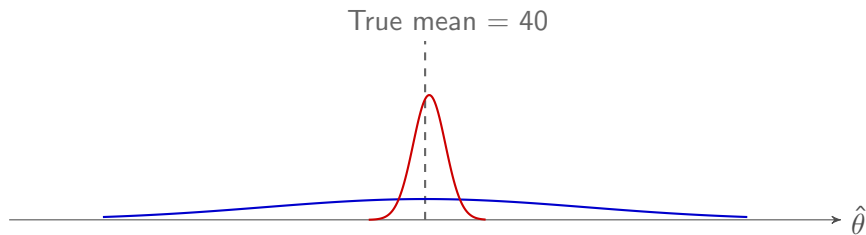
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- $\text{MSE}(\hat{\theta}_2) = \text{Var}(\hat{\theta}_2) + \text{Bias}(\hat{\theta}_2)^2 = \frac{\sigma^2}{1000} + (0.1)^2 = 0.16.$

Graphical Illustration of the Tradeoff

The distribution of $\hat{\theta}_1$ is centered correctly but very wide. The distribution of $\hat{\theta}_2$ is slightly off-center but extremely narrow. An estimate from $\hat{\theta}_2$ is almost always closer to the true value μ .



Maximum Likelihood Estimation (MLE)

Idea: Choose the parameter $\hat{\theta}$ that *maximizes* the likelihood of the observed data. The likelihood function:

$$L(\theta) = \prod_{i=1}^n f(x_i | \theta)$$

We maximize this, or more commonly, the log-likelihood:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i | \theta)$$

Workflow

1. Write down the log-likelihood function $\ell(\theta)$.
2. Differentiate to get the score function $\frac{\partial \ell}{\partial \theta}$.
3. Set the score function to zero and solve for θ .

Key Properties of MLE (Intuition)

Under general "regularity conditions," Maximum Likelihood Estimators have excellent properties:

- ▶ **Consistency:** As $n \rightarrow \infty$, $\hat{\theta}_{\text{MLE}}$ converges in probability to the true θ . (We saw this graphically!)
- ▶ **Asymptotic Normality:** For large n , the distribution of $\hat{\theta}_{\text{MLE}}$ is approximately Normal.

$$\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, I_1(\theta)^{-1})$$

- ▶ **Asymptotic Efficiency:** For large n , MLEs achieve the Cramér-Rao Lower Bound. They are as good as any unbiased estimator can be.
- ▶ **Invariance:** For a function g , the MLE of $g(\theta)$ is simply $g(\hat{\theta}_{\text{MLE}})$. For example, the MLE of p^2 is $\hat{p}^2$.

Bernoulli(p)

Data: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$, $x_i \in \{0, 1\}$. Let $S = \sum x_i$.

$$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^S (1-p)^{n-S},$$

$$\ell(p) = S \log p + (n-S) \log(1-p).$$

The score equation is:

$$\frac{d\ell}{dp} = \frac{S}{p} - \frac{n-S}{1-p} = 0 \quad \Rightarrow \quad \hat{p} = \frac{S}{n} = \bar{X}.$$

Properties:

- ▶ $\mathbb{E}[\hat{p}] = p$ (Unbiased)
- ▶ $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$
- ▶ $I_n(p) = \frac{n}{p(1-p)}$, so $\text{CRLB} = \frac{p(1-p)}{n}$. (Efficient)

Poisson(λ)

Data: $X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$. The PMF is $f(x_i|\lambda) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$.

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$$\ell(\lambda) = \sum_{i=1}^n (x_i \log \lambda - \lambda - \log x_i!)$$

$$\frac{d\ell}{d\lambda} = \sum_{i=1}^n \left(\frac{x_i}{\lambda} - 1 \right) = \frac{\sum x_i}{\lambda} - n = 0$$

Solving gives the MLE:

$$\hat{\lambda} = \frac{\sum x_i}{n} = \bar{X}$$

Properties: $\mathbb{E}[\bar{X}] = \lambda$, $\text{Var}(\bar{X}) = \frac{\lambda}{n}$, $l_1(\lambda) = \frac{1}{\lambda}$, $\text{CRLB} = \frac{\lambda}{n}$. The MLE is efficient.

Normal with Known Variance

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Properties: $\mathbb{E}[\hat{\mu}] = \mu$, $\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$, $h_1(\mu) = \frac{1}{\sigma^2}$, $\text{CRLB} = \frac{\sigma^2}{n}$.
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Notes:

- ▶ $\hat{\sigma}_{\text{MLE}}^2$ is *biased*; $\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \frac{n-1}{n}\sigma^2$. The unbiased version uses a denominator of $n-1$ (the sample variance S^2).
- ▶ By invariance, the MLE for the standard deviation is $\hat{\sigma}_{\text{MLE}} = \sqrt{\hat{\sigma}_{\text{MLE}}^2}$.

Exponential(λ) with Rate Parameter

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Bias: This estimator is biased. $\mathbb{E}[\hat{\lambda}] = \frac{n}{n-1}\lambda$ for $n > 1$.

Information: $I_1(\lambda) = \frac{1}{\lambda^2}$, so $I_n(\lambda) = \frac{n}{\lambda^2}$.

Uniform($0, \theta$)

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maximize $L(\theta)$, we need to make θ as small as possible. However, θ must be greater than or equal to all observations. Therefore, the MLE is the smallest possible value that is still valid:

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Bias: $\mathbb{E}[\hat{\theta}] = \frac{n}{n+1}\theta$; it is biased but consistent.

Applications of MLE in AI

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- ▶ **Generative Models (VAEs):** Variational Autoencoders are deep generative models trained to estimate the probability distribution of the data. Their objective function is designed as a tractable lower bound to the true data log-likelihood.

Summary

- ▶ Estimators are evaluated based on their **bias**, **variance**, and **MSE**. There is often a tradeoff between bias and variance.
- ▶ The **CRLB** provides a theoretical benchmark for the lowest possible variance of any unbiased estimator.
- ▶ **Maximum Likelihood Estimation** is a powerful and principled method for deriving estimators.
- ▶ MLEs are generally **consistent**, **asymptotically normal**, and **asymptotically efficient**.
- ▶ We've seen how to derive and analyze MLEs for key distributions like the Bernoulli, Poisson, Normal, Exponential, and Uniform.